

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4.	(a)	(i)		$2\,000 \times 0.15 = [\pounds] 300$		1		1	1	
		(ii)		Substitution: $\frac{1800}{300}$ ecf (1) = 6 [years] (1)	1	1		2	2	
	(b)	(i)		Wind generator 2 has a payback time of 5 years (1) but wind generator 3 has a payback time of 6.5 years (1) [so salesperson is not correct.] Award 2 marks for the payback time for wind generator 3 is 1.5 years longer. Alternative: Wind generator 2 costs £75 more (1) But saves £150 more each year or saves £750 more over 5 years (1) [so salesperson is not correct.] Or Wind generator 3 costs £75 less (1) But saves £150 less each year or saves £750 less over 5 years (1) [so salesperson is not correct.] Or Wind generator 2 costs £75 more (1) But saves £3 000 in 5 years compared to £2250 (1) [so salesperson is not correct.] Treat as neutral any comparison with wind generator 1			2	2		
		(ii)		<u>Cost</u> of kWh / unit may change (1) accept price of electricity can change Wind conditions are variable (1) accept weather changes Accept for 1 mark maximum: more or less electricity may be used Don't accept cost will change or number of units produced will change		2		2		

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					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		B → C → A All three letters in correct order award 2 marks One or two letters correct award 1 mark		2		2		2
		(ii)	I	X		1		1	1	1
			II	Z		1		1	1	1
	(d)	(i)		Stronger magnet / move magnet closer to the coil / lighter magnet Don't accept add another magnet or smaller magnet			1	1		1
		(ii)		More <u>turns</u> . Accept more coils.			1	1		1
				Question 4 total	1	8	4	13	5	6